Pratik Panwar | BS22B029
Indian Institute of Technology Madras

PratikPanwar Pratik464iitm

Education

- **Indian Institute of Technology Madras** *Nov'22–May'27*
  Dual Degree in Biotechnology | CGPA: **7.98/10**

- **Shri Kanwartara Public Hr. Sec. School, Mandleshwar**
  Class 12 (CBSE): **92.6%** *May'21–Mar'22*

- **Shri Kanwartara Public Hr. Sec. School, Mandleshwar**
  Class 10 (CBSE): **91.0%** *May'19–Mar'20*

Scholastic Achievements

- Qualified for **National Talent Search Examination (NTSE)** Stage 2, placing in the **top 0.1%** among **1M+** candidates*(2020)*
- Qualified **JEE Main** among **1M+** candidates and **JEE Advanced** among **250K+** candidates across India *(2022)*

Professional Experience

Soil Sarthi
Full-Stack Developer *Mar'25–May'25*

- Built a full-stack AgriTech platform using **React.js, Node.js, Express.js,** and **MongoDB** to enable digital farming services
- Developed **RESTful APIs** and designed **MongoDB schemas** with Mongoose, testing and validating endpoints via **Postman**
- Automated publishing of **30+ blog posts** through the **LinkedIn API**, reducing manual content publishing effort
- Built a voice-based farmer query feature using the **OpenAI Whisper API** to convert spoken agricultural queries into text, enabling seamless and accessible platform interaction

Coding Achievements

- Secured a **PPI at InMobi** by ranking among the **top performers** in the **InMobi Codies 2026 Coding Contest**
- Achieved **Global Rank 488** among **36K+** participants in the **ICPC 2025 Online Winter Challenge** powered by Huawei
- Secured **Rank 416** among **14K+** participants in the **Goldman Sachs India Hackathon 2026 – CS** challenge

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- Achieved a maximum **Codeforces** rating of **1667 (Expert)**
- Secured **Global Rank 386** in Codeforces Round 1102 (Div. 2) and **Global Rank 496** in Order Capital Round 2 (Div. 1 + 2)
- Achieved **Global Rank 101** in Starters 241 (CodeChef)
- Rated **1809 (Knight)** on **LeetCode**, placing in the **top 7.75%**

- Ranked **18th** in **Reverse Coding Contest** (250+ participants)
- Ranked **11th** in **Heuristics Contest** held by Programming Club
- Ranked **12th** in **DSA Hackathon** held by Programming Club

Skills

Programming Languages: C++, C, Python, JavaScript, SQL
Web Dev: HTML, CSS, React.js, Node.js, Express.js, Socket.IO
Databases & Dev. Tools: MongoDB, Git, GitHub, Postman

Courses

- Data Structures & Algorithms
- Operating Systems*
- Computer Networks*
- Machine Learning Foundations
- Programming in Python
- Principles of Economics
- Web Development
- Object Oriented Programming
- Series and Matrices
- Money, Banking & Financial Markets

* : Ongoing

Projects

SyncBoard
Real-Time Collaborative Workspace *MERN, JWT, Socket.IO*

- Developed a **MERN-based** collaborative workspace with **RESTful APIs** for Workspaces, Boards, Columns, and Tasks, implementing complete **CRUD** functionality across modules
- Implemented **JWT authentication** and **RBAC** with workspace-level permissions for secure member collaboration
- Integrated **Socket.IO** for real-time synchronization and implemented **search, pagination, validation,** and **centralized error handling** across the entire application

HuffZip
Lossless File Compression Utility *C++, STL, OOP, File I/O*

- Developed a **Huffman Coding**–based file compression system for **lossless compression** and decompression of text files using a **greedy algorithm** and **binary tree encoding**
- Implemented **Huffman Tree construction, code generation, bit packing,** and **binary file I/O** for efficient compression
- Designed a modular compression/decompression pipeline with separate encoding, decoding, and file-handling components

ChatRelay
Real-Time C++ Messaging Engine *Multithreading, Boost.Asio*

- Developed a **C++ client-server chat application** supporting real-time messaging among multiple users over **TCP sockets**
- Implemented **multithreaded, asynchronous client handling** using **Boost.Asio**, enabling concurrent user connections and efficient non-blocking message handling
- Designed a **structured messaging protocol** supporting private messaging, user renaming, and active client tracking for efficient message routing among connected clients

AlgoViz
Algorithm Visualization Platform *React, JavaScript, DSA*

- Developed an interactive algorithm visualization platform demonstrating **sorting**, **graph**, **and searching algorithms**
- Implemented **Bubble**, **Selection**, **Insertion**, **Merge**, **and Quick Sort** using a reusable dynamic event-driven animation engine
- Integrated **BFS**, **DFS**, **and Dijkstra's Algorithm** to visualize graph traversal and shortest-path computation step by step

Positions of Responsibility

Alumni Relations Cell
Strategist *Apr'24–May'25*

- Led **Recall**, ARC's flagship mentorship initiative, hosting **10 alumni sessions** and managing a budget of **₹5+ lakhs**
- Coordinated an internship program achieving **100% placement**, facilitating **12 hires** from a pool of **500+** applicants

Teaching Assistant
GN6106 *May'23–Mar'24*

- Assisted with course coordination and academic activities for **90+ students** under **Prof. Sridhar Boovaraghavan**

Extra-Curriculars

Sports

- Selected for the **20-member Cricket NSO Contingent** at IIT Madras from a competitive pool of **200+** applicants
- Part of the **IIT Madras Inter-IIT Cricket** probables
- Represented Brahmaputra (2024) and Godavari (2025–26) in the institute's **Inter-Hostel Cricket League** (Schorters)
- Secured **2nd Runner-up** in the Inter-Departmental Cricket League and was named **2nd Most Valuable Player**

- **SAATHI Mentor:** Mentored 3 freshmen, supporting their transition to college through peer engagement